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Outcomes After Transcatheter Aortic Valve Replacement in Bicuspid Versus Tricuspid Anatomy

A Systematic Review and Meta-Analysis

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ABSTRACT

OBJECTIVES The aim of this study was to compare the feasibility, safety, and clinical outcomes of transcatheter aortic valve replacement (TAVR) in bicuspid aortic valve (BAV) versus tricuspid aortic valve (TAV) stenosis.

BACKGROUND At present, limited observational data exist supporting TAVR in the context of bicuspid anatomy.

METHODS Primary endpoints were 1-year survival and device success. Secondary endpoints included moderate to severe paravalvular leak (PVL) and a composite endpoint of periprocedural complications; incidence rates of individual procedural endpoints were also explored individually.

RESULTS In the main analysis, 17 studies and 181,433 patients undergoing TAVR were included, of whom 6,669 (0.27%) had BAV. A secondary analysis of 7,071 matched subjects with similar baseline characteristics was also performed. Device success and 1-year survival rates were similar between subjects with BAV and those with TAV (97% vs 94% [$P = 0.55$] and 91.3% vs 90.8% [$P = 0.22$], respectively). In patients with BAV, a trend toward a higher risk for periprocedural complications was observed in our main analysis (risk ratio [RR]: 1.12; 95% CI: 0.99-1.27; $P = 0.07$) but not in the matched population secondary analysis (RR: 1.00; 95% CI: 0.81-1.24; $P = 0.99$). The risk for moderate to severe PVL was higher in subjects with BAV (RR: 1.42; 95% CI: 1.29-1.58; $P < 0.0001$) as well as the incidence of cerebral ischemic events (2.4% vs 1.6%; $P = 0.015$) and of annular rupture (0.3% vs 0.02%; $P = 0.014$) in matched subjects.

CONCLUSIONS TAVR is a feasible option among selected patients with BAV anatomy, but the higher rates of moderate to severe PVL, annular rupture, and cerebral ischemic events observed in the BAV group warrant caution and further evidence. (J Am Coll Cardiol Intv 2021;14:2144–2155) © 2021 by the American College of Cardiology Foundation.

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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Bicuspid aortic valve (BAV) is the most common congenital heart disease (present in 1%-2% of the population) (1) and carries a higher risk for progression to severe aortic stenosis (2) even in younger patients (3). Transcatheter aortic valve replacement (TAVR) has shown comparable outcomes as surgery for severe aortic stenosis even in patients at low surgical risk (4,5). However, BAV poses specific challenges for performing TAVR due to anatomical features of the bicuspid anatomy. At present, limited observational data exist supporting TAVR in the context of bicuspid anatomy. In this systematic review and meta-analysis, we sought to compare the feasibility, safety, and outcomes of TAVR in BAV versus tricuspid aortic valve (TAV). Furthermore, we explored what impact different device generations and valves can have on these outcomes.

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METHODS

STUDY SELECTION. Two authors (C.M., A.S.) independently searched PubMed, Embase, BioMed Central, Google Scholar, and the Cochrane Central Register of Controlled Trials for reports published between January 1, 2000, and April 1, 2020; the full search strategy appears in the Supplemental Methods. Papers were initially screened by title and abstract content. In addition, the reference lists of all eligible reports were screened to identify any additional citations. Studies in which clinical outcomes of adult patients undergoing TAVR with BAV or TAV were compared were included. In some cases, in which the risk for overlap between cohorts was considered to be high, only the larger reported cohorts were included in the analysis. In 2 cases (6,7), the studies excluded for this reason reported outcomes of matched populations and were therefore included in the dedicated secondary analysis. All controversies were discussed and adjudicated by a senior author (A.M.). The present work was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses and Meta-Analysis of Observational Studies in Epidemiology guidelines (8,9). The study protocol has been submitted to PROSPERO (CRD42020185707) (10). Two investigators (C.M., A.S.) performed data extraction independently, and the extracted data were verified by a third investigator (G.C.) for accuracy.

ELIGIBILITY CRITERIA. Studies were considered eligible if they fulfilled all of the following criteria: 1) they reported outcomes of patients with BAV and TAV undergoing TAVR; and 2) they reported at least 1 of our primary endpoints (survival rates or device success).

Case reports, case series, letters, and studies that failed to clearly report the numbers and rates of patients alive at follow-up were excluded from the analysis. Two authors (C.M., A.S.) independently assessed the quality of studies and risk for bias according to the ROBINS-I tool (11). All studies included had appropriate ethical oversight and approval.

ABBREVIATIONS AND ACRONYMS

BAV = bicuspid aortic valve

PVL = paravalvular leak

RR = risk ratio

TAV = tricuspid aortic valve

TAVR = transcatheter aortic valve replacement

TIA = transient ischemic attack

OUTCOMES AND EFFECT SUMMARY. The primary endpoints of this analysis were device success and survival at 1 year.

Secondary endpoints were: 1) periprocedural complications, defined as the composite of major vascular complications, myocardial infarction, stroke or transient ischemic attack (TIA), major or life-threatening bleeding, new pacemaker implantation, and acute kidney injury; 2) moderate to severe paravalvular leak (PVL); and 3) incidence rates of stroke or TIA, annular rupture, coronary occlusion, device embolization, conversion to surgery, and need for a second valve.

A further subgroup analysis was performed to evaluate the impact of transcatheter valve type and device generation on the aforementioned endpoints. The definition of device success was consistent with the Valve Academic Research Consortium definition (12) in the majority of studies (**Table 1**).

META-ANALYSIS. Pooled risk ratios (RRs) and standardized mean differences with 95% CIs were used as summary statistics for outcomes of interest and were calculated using a random-effects model according to DerSimonian and Laird (13). The baseline characteristics were pooled according to aortic valve anatomy (BAV vs TAV) and presented as pooled weighted means and 95% CIs with *P* values for differences between groups. In a secondary analysis, only studies reporting outcomes from matched populations were included. Statistical heterogeneity of exposure was assessed by calculating the I^2 index, which summarizes the amount of variance among studies beyond chance. Heterogeneity was considered to be low for I^2 <25%, medium for I^2 <75%, and high for I^2 >75% (14,15). Because the I^2 statistic was medium to high in our main analysis, the random effect was considered the primary model. To account for heterogeneity in follow-up, we calculated linearized survival rates with the person-year method and pooled outcomes on a logarithmic scale in a random-effect analysis using the inverse variance method. Residual heterogeneity of primary endpoints was further assessed by a random-effect

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TABLE 1 Characteristics of Studies Included in the Analysis

First Author	Journal, Year	Study Design	Enrollment Period	FU	BAV	TAV	Matched BAV	Matched TAV	New-Generation Valve	Old-Generation Valve	Device Success Definition
Hayashida et al	*Circulation: Cardiovascular Intervention*, 2013	Single center, retrospective	2006-2012	30 d	21	208	–	–	–	Edwards SAPIEN XT, Medtronic CoreValve	VARC
Bauer et al	*American Journal of Cardiology*, 2014	Multicenter, prospective registry	2009-2010	1 y	38	1,357	–	–	–	Edwards SAPIEN XT, Medtronic CoreValve	VARC[a]
Costopoulos et al	*American Journal of Cardiology*, 2014	Multicenter, retrospective	2007-2012	1 y	21	447	–	–	–	Edwards SAPIEN, Medtronic CoreValve	VARC-2
Kochman et al	*American Journal of Cardiology*, 2014	Multicenter retrospective	2009-2012	1 y	28	84	–	–	–	Edwards SAPIEN XT, Medtronic CoreValve	VARC
Liu et al	*Journal of Zhejiang University-Science B*, 2015	Single center, retrospective	2013-2014	30 d	–	–	15	35	–	Medtronic CoreValve, Venus-A	VARC
Sannino et al	*American Journal of Cardiology*, 2017	Multicenter, retrospective	2012-2016	1 y	88	735	–	–	Edwards SAPIEN 3, Medtronic CoreValve Evolut R, Boston Lotus	Edwards SAPIEN XT, Medtronic CoreValve	VARC-2
Yoon et al	*Journal of the American College of Cardiology*, 2017	Multicenter, international, retrospective	2005-2016	2 y[a]	561	4,546	546	546	Edwards SAPIEN 3, Medtronic CoreValve Evolut R, Boston Lotus	Edwards SAPIEN XT, Medtronic CoreValve	VARC-2
Kawamori et al	*European Heart Journal: Cardiovascular Imaging*, 2017	Single center, retrospective analysis of one RCT and one prospective registry	2013-2017	30 d	41	239	–	–	Edwards SAPIEN 3	–	VARC-2
Liao et al	*International Journal of Cardiology*, 2017	Single center, prospective	2012-2017	1 y	87	70	–	–	–	Medtronic CoreValve, Venus-A	VARC-2
Mangieri et al	*EuroIntervention*, 2018	Multicenter, international, retrospective	2012-2017	30 d	54	658	54	54	Boston ACURATE neo	–	VARC-2
Xiong et al	*International Journal of Cardiovascular Imaging*, 2018	Single center, retrospective	2012-2016	1 y	67	49	–	–		Medtronic CoreValve, Venus-A, MicroPort	VARC-2
De Biase et al[b]	*International Journal of Cardiology*, 2018	Single center, retrospective	2016-2017	30 d	–	–	83	166	Edwards SAPIEN 3, Medtronic CoreValve Evolut R, Boston Lotus, St Jude Portico	-	VARC-2
Aalaei-Andabili et al	*Journal of Cardiac Surgery*, 2018	Single center, retrospective	2012-2017	2 y	–	–	32	96	Edwards SAPIEN 3	Edwards SAPIEN XT, Medtronic CoreValve	VARC
Tchetche et al	*Circulation: Cardiovascular Intervention*, 2019	Multicenter, international, retrospective registry	2016-2017	30 d	101	88	–	–	Edwards SAPIEN 3, Medtronic CoreValve Evolut R, Boston Lotus		VARC-2
Xu et al	*Herz*, 2019	Multicenter, prospective trial		1 y	9	14	–	–	Boston Lotus		VARC-2
Makkar et al[b]	*JAMA*, 2019	Multicenter, prospective registry	2014-2018	1 y	–	–	2,691	2,691	Edwards SAPIEN 3	–	VARC-2

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subgroup analysis according to differences in valve type (self- vs balloon-expandable) and device generation. Statistical significance was set at $P < 0.05$ (2 sided). Finally, sensitivity analyses were performed for primary endpoints by assessing the effect of removing individual studies on the pooled RR. Publication bias was assessed for primary endpoints by visual inspection of funnel plots and by

TABLE 1 Continued

First Author	Journal, Year	Study Design	Enrollment Period	FU	BAV	TAV	Matched BAV	Matched TAV	New-Generation Valve	Old-Generation Valve	Device Success Definition
Halim et al	*Circulation*, 2019	Multicenter, prospective registry	2011-2018	1 y	5,412	165,547	–	–	Edwards SAPIEN 3, Medtronic CoreValve Evolut R	Edwards SAPIEN XT, Medtronic CoreValve	VARC-2
Bo Fu et al	*Annals of Translational Medicine*, 2020	Single center, retrospective	2019	1 y	44	74	–	–	–	Venus-A	VARC-2
Pineda et al	*American Heart Journal*, 2020	Single center, retrospective	2011-2016	2 y	50	517	–	–	Edwards SAPIEN 3, Medtronic Corevalve Evolut R	Edwards SAPIEN XT Medtronic CoreValve	NA

Values are n, unless otherwise indicated. [a]Considered VARC equivalent by consensus. The text defines valve success as "deployment of the valve as intended, without need for conversion to immediate surgery." [b]Studies included in the matched-cohort secondary analysis only.

BAV = bicuspid aortic valve; FU = follow-up; NA = not applicable; TAV = tricuspid aortic valve; VARC = Valve Academic Research Consortium.

Egger's and Begg's tests. Data analysis was performed in the R environment (packages meta and metafor) (R Foundation for Statistical Computing).

RESULTS

SYSTEMATIC REVIEW OF PUBLISHED STUDIES. Our search identified a total of 19 studies; the process of study selection is summarized in **Figure 1**. Seventeen studies (16-32) were included in this systematic review, with a total of 181,433 patients undergoing TAVR, 174,764 (99.73%) with TAV and 6,669 (0.27%) with BAV. All studies were interventional and clearly reported follow-up times, ranging from 30 days to 2 years (**Table 1**), with an overall moderate risk for bias (Supplemental Table 1). Only 6 studies reported outcomes of matched populations, for a total of 7,071 subjects undergoing TAVR (3,434 and 3,637 with BAV and TAV, respectively). Patients were treated with different devices, including both old- and new-generation self- and balloon-expandable valves. Old-generation devices were Edwards SAPIEN and SAPIEN XT, Medtronic CoreValve, Boston Scientific Lotus, Venus-A, and MicroPort VitaFlow; all other devices were considered new generation. Additional information on the individual studies is reported in **Table 1**. Pooled baseline characteristics of the patients included are presented in **Table 2**. In the unmatched cohorts, patients with BAV were significantly younger (76.1 years vs 80.3 years; $P < 0.0001$), with lower left ventricular ejection fractions (52.7% vs 55.5%; $P = 0.030$), and were more frequently in New York Heart Association functional class III or IV (78% vs 65%; $P = 0.020$). In the unmatched cohorts, patients with BAV were more frequently treated with new-generation devices (46% vs 27%; $P = 0.03$), without any other significant difference. No publication bias was identified by visually inspecting funnel plots and by mathematical testing (Supplemental Figure 1). Sensitivity analysis were performed by excluding each study individually from the analysis, with no relevant changes to primary outcomes. (Supplemental Table 1) Risk for bias assessment is reported in Supplemental Methods.

PRIMARY OUTCOMES: 1-YEAR SURVIVAL AND DEVICE SUCCESS. Our main analysis showed that 1-year mortality and device success after TAVR were similar in BAV and TAV patients (RR: 1.10; 95% CI: 0.84-1.45; $P = 0.45$; and RR: 1.01; 95% CI: 0.98-1.03; $P = 0.53$, respectively), which was also confirmed in our secondary analysis of matched patients (RR: 0.91; 95% CI: 0.77-1.06; $P = 0.23$; and RR: 0.96; 95% CI: 0.91-1.01; $P = 0.22$) (**Figure 2**). Our subgroup analysis revealed that neither device generation nor valve type influenced device success. A borderline-significant trend toward lower occurrence of all-cause death at 1 year was observed in BAV subjects treated with newer generation devices (RR: 0.84; 95% CI: 0.70-1.0; $P = 0.05$).

SECONDARY OUTCOMES: PERIPROCEDURAL COMPLICATIONS AND PVL. A trend toward a higher risk for periprocedural complications in patients with BAV was observed in our main analysis (RR: 1.12; 95% CI: 0.99-1.27; $P = 0.07$) but not in the matched population secondary analysis (RR: 1.00; 95% CI: 0.81-1.24; $P = 0.99$) (**Figure 2**); forest plots for individual components are shown in Supplemental Figure 2. The occurrence of moderate to severe PVL was significantly higher in the BAV cohort in the main analysis (RR: 1.42; 95% CI: 1.29-1.58; $P < 0.0001$) and in the matched populations (RR: 1.53; 95% CI: 1.17-2.03; $P < 0.002$) (**Figure 3**). The incidence rates of stroke or TIA and of annular rupture were higher in BAV than TAV (**Table 3**), with a statistically significant difference in the matched population ($P = 0.015$ and

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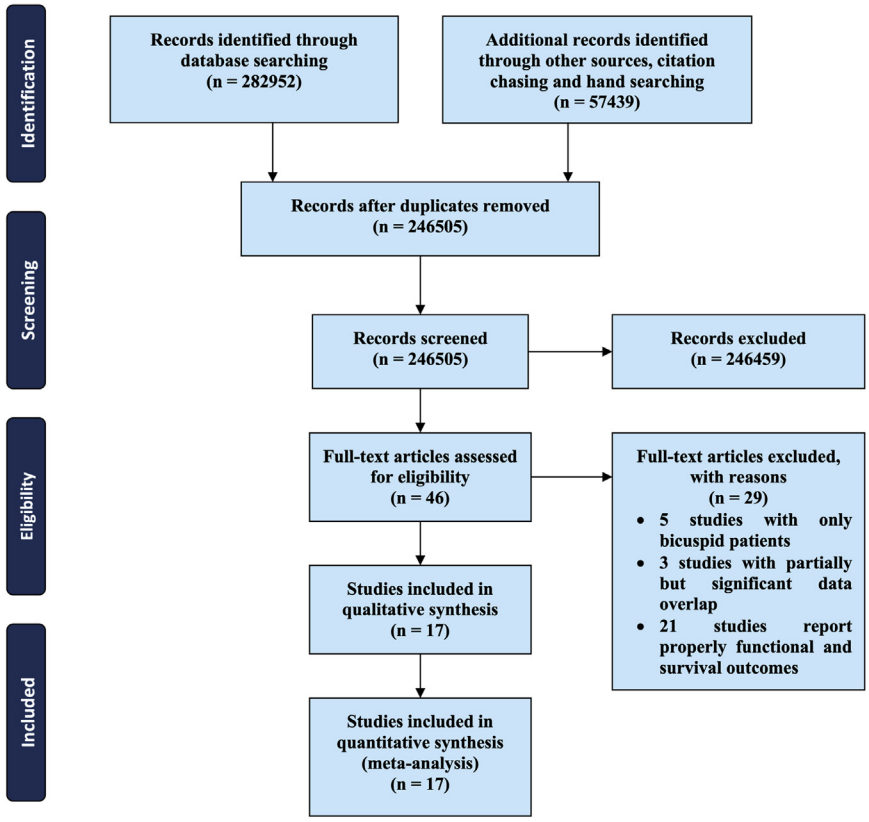
FIGURE 1 PRISMA Flowchart



PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

$P = 0.014$, respectively). The occurrence of individual complications, such as the need for a second valve ($P = 0.007$) and conversion to surgery ($P = 0.018$), was higher in BAV subjects.

SUBGROUP ANALYSIS AND META-REGRESSION. In our subgroup analysis, the RR of periprocedural complications was significantly higher in subjects with BAV when new-generation devices and balloon-expandable valves were used ($P = 0.01$ and $P = 0.009$, respectively). Moreover, BAV was associated with increased occurrence of moderate to severe PVL when self-expandable devices were used, but irrespective of device generation. In contrast, the use of a balloon-expandable valve was associated with similar rates of moderate to severe PVL irrespective of aortic valve anatomy (RR: 1.11; 95% CI: 0.62-1.97; $P = 0.72$).

Our meta-regression analysis revealed that neither follow-up length nor baseline parameters (including Society of Thoracic Surgeons score, age, left ventricular ejection fraction, and advanced New York Heart Association functional class) significantly affected our primary endpoints. Also, there was no significant relationship between year of publication and our primary endpoints (Supplemental Figures 3 and 4).

DISCUSSION

The main findings of our large meta-analysis of 181,433 subjects undergoing TAVR are as follows (**Central Illustration**): 1) overall device success and 1-year mortality were equivalent following TAVR in BAV or TAV anatomy; 2) TAVR in BAV resulted in a higher rate of residual moderate to severe PVL compared with TAVR in TAV, but this difference was mitigated with the use of balloon-expandable valves; 3) a similar rate of periprocedural complications was observed in BAV and in TAV in the matched population; and 4) the rate of periprocedural complications was higher when new-generation valves and balloon-expandable valves were implanted; nevertheless, the use of balloon-expandable valves mitigates the risk for residual significant PVL, while self-expandable

TABLE 2 Baseline Characteristics of Patients Included in the Analysis

	Unmatched Population (n = 181,433)				Matched Population (n = 7,071)			
	Number of Studies	BAV	TAV	*P* Value	Number of Studies	BAV	TAV	*P* Value
Age (y)	17	76.0 (74.69-77.47)	80.2 (79.76-80.81)	<0.001	6	76.7 (74.20-79.32)	78.0 (74.74-81.37)	0.5445
Female (%)	17	0.43 (0.39-0.47)	0.38 (0.34-0.44)	0.2064	6	0.40 (0.35-0.46)	0.39 (0.36-0.41)	0.6708
BMI (kg/m^2)	10	24.6 (23.21-26.07)	24.8 (23.65-26.10)	0.8073	3	29.98 (26.70-33.26)	30.0 (26.85-33.23)	0.9788
STS score	13	6.29 (5.28-7.30)	7.04 (6.44-7.63)	0.2118	5	4.87 (4.61-5.13)	5.00 (4.50-5.49)	0.6496
LVEF (%)	14	52.6 (50.99-54.30)	55.4 (53.52-57.44)	0.0303	6	53.16 (51.39-54.93)	52.2 (51.80-52.77)	0.3486
Mean gradient (mm Hg)	14	51.9 (48.47-55.46)	48.8 (47.03-50.75)	0.1277	6	47.93 (45.21-50.64)	46.4 (44.31-48.60)	0.4048
AVA (cm^2)	13	0.63 (0.6070-0.6663)	0.66 (0.6427-0.6789)	0.1727	5	0.65 (0.6167-0.6878)	0.67 (0.6539-0.7001)	0.2534
CAD (%)	9	0.47 (0.3965-0.5667)	0.59 (0.5414-0.6648)	0.0250	2	0.47 (0.3649-0.6252)	0.55 (0.4231-0.7250)	0.4462
Prior MI (%)	10	0.10 (0.0555-0.1862)	0.15 (0.1060-0.2145)	0.2699	4	0.14 (0.0554-0.3674)	0.15 (0.0676-0.3599)	0.8898
Prior PCI (%)	12	0.19 (0.1386-0.2769)	0.27 (0.2326-0.3252)	0.0836	5	0.25 (0.2173-0.2884)	0.29 (0.2468-0.3578)	0.1501
Prior CABG (%)	8	0.09 (0.0649-0.1362)	0.10 (0.0601-0.1775)	0.7787	5	0.12 (0.0907-0.1675)	0.13 (0.0905-0.1893)	0.8069
Prior stroke (%)	15	0.15 (0.0922-0.2647)	0.13 (0.1153-0.1510)	0.5428	5	0.23 (0.0858-0.6546)	0.11 (0.0978-0.1352)	0.1683
PAD (%)	15	0.23 (0.1865-0.2855)	0.26 (0.2309-0.3001)	0.3018	5	0.18 (0.1303-0.2556)	0.22 (0.1652-0.2945)	0.4034
Diabetes (%)	16	0.26 (0.1278-0.5530)	0.30 (0.2776-0.3388)	0.7045	6	0.29 (0.2247-0.3832)	0.29 (0.2268-0.3852)	0.9700
Hypertension (%)	14	0.65 (0.5747-0.7468)	0.77 (0.7231-0.8265)	0.0273	6	0.77 (0.6849-0.8669)	0.76 (0.6927-0.8523)	0.9720
COPD (%)	16	0.29 (0.2420-0.3711)	0.28 (0.2518-0.3129)	0.5927	5	0.28 (0.1791-0.4623)	0.26 (0.1592-0.4255)	0.7738
CKD (%)	14	0.21 (0.0984-0.4593)	0.22 (0.1002-0.5176)	0.9051	4	0.10 (0.0204-0.5611)	0.14 (0.0416-0.5309)	0.7588
NYHA functional class III or IV (%)	15	0.78 (0.7294-0.8359)	0.65 (0.5771-0.7473)	0.0202	5	0.75 (0.6930-0.8159)	0.72 (0.6589-0.8044)	0.6236
Moderate to severe MR (%)	6	0.21 (0.1552-0.3008)	0.23 (0.1746-0.3285)	0.6588	2	0.19 (0.1140-0.3248)	0.24 (0.1237-0.4851)	0.5826
Moderate to severe AR (%)	8	0.11 (0.0487-0.2939)	0.20 (0.0718-0.5607)	0.4574	na			
Transfemoral TAVR (%)	17	0.89 (0.8731-0.9185)	0.85 (0.8092-0.8952)	0.0779	6	0.87 (0.7918-0.9677)	0.86 (0.7703-0.9611)	0.8220
New-generation device (%)	16	0.46 (0.3466-0.6260)	0.27 (0.1804-0.4127)	0.0393	5	0.52 (0.3101-0.8839)	0.54 (0.3187-0.9394)	0.9084
Self-expanding device (%)	17	0.53 (0.3354-0.8478)	0.44 (0.2753-0.7038)	0.5688	5	0.45 (0.2620-0.7759)	0.50 (0.3208-0.7869)	0.7630
Post-TAVR mean gradient (mm Hg)	15	10.5 (9.75-11.36)	9.77 (9.16-10.37)	0.1241	6	10.56 (9.74-11.38)	10.31 (9.36-11.25)	0.6960
Post-TAVR AVA (cm^2)	5	1.78 (1.62-1.95)	1.80 (1.69-1.90)	0.86	2	1.67 (1.42-1.93)	1.70 (1.51-1.89)	0.8517

Values are pooled value (95% CI).

AR = aortic regurgitation; AVA = aortic valve area; BMI = body mass index; CABG = coronary artery bypass grafting; CAD = coronary artery disease; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; LVEF = left ventricular ejection fraction; MI = myocardial infarction; MR = mitral regurgitation; NYHA = New York Heart Association; PAD = peripheral artery disease; PCI = percutaneous coronary intervention; STS = Society of Thoracic Surgeons; TAVR = transcatheter aortic valve replacement; other abbreviations as in Table 1.

valves convey a higher risk for moderate to severe PVL when implanted in bicuspid anatomies compared with TAV.

BAV disease is anatomically more challenging to treat percutaneously than conventional TAV stenosis. The often larger and more elliptical annulus with a calcified raphe can hinder adequate symmetrical expansion of the transcatheter heart valve, while the volcano-shaped anatomy (33) alters the valve sealing and anchoring process. These challenges explain the rationale behind the exclusion of BAV disease in the pivotal studies, which cemented the benefit of TAVR over surgery with the use of balloon- and self-expandable valves (4,5).

Moreover, as TAVR indications expand toward lower risk and younger subjects, the occurrence of severe aortic stenosis in a bicuspid anatomy is likely to increase. However, the current evidence base supporting the use of TAVR in BAV is weak, as no data from randomized clinical trials exist, and published research is limited to observational registries that seldom report separate outcomes with regard to device type or generation.

Prior studies using older generation devices suggested that TAVR in bicuspid anatomy was associated with lower device success (85.3% vs 91.4%; $P = 0.002$) (28) and confirmed in a recent meta-analysis of unselected subjects (34). However, recent multicenter registries reported more favorable outcomes with the use of newer generation devices (Medtronic CoreValve Evolut R/PRO and Edwards SAPIEN 3), albeit in relatively small cohorts (35,36).

Our study showed similar rates of device success (97% and 94%; $P = 0.55$) and 1-year mortality (91.3%

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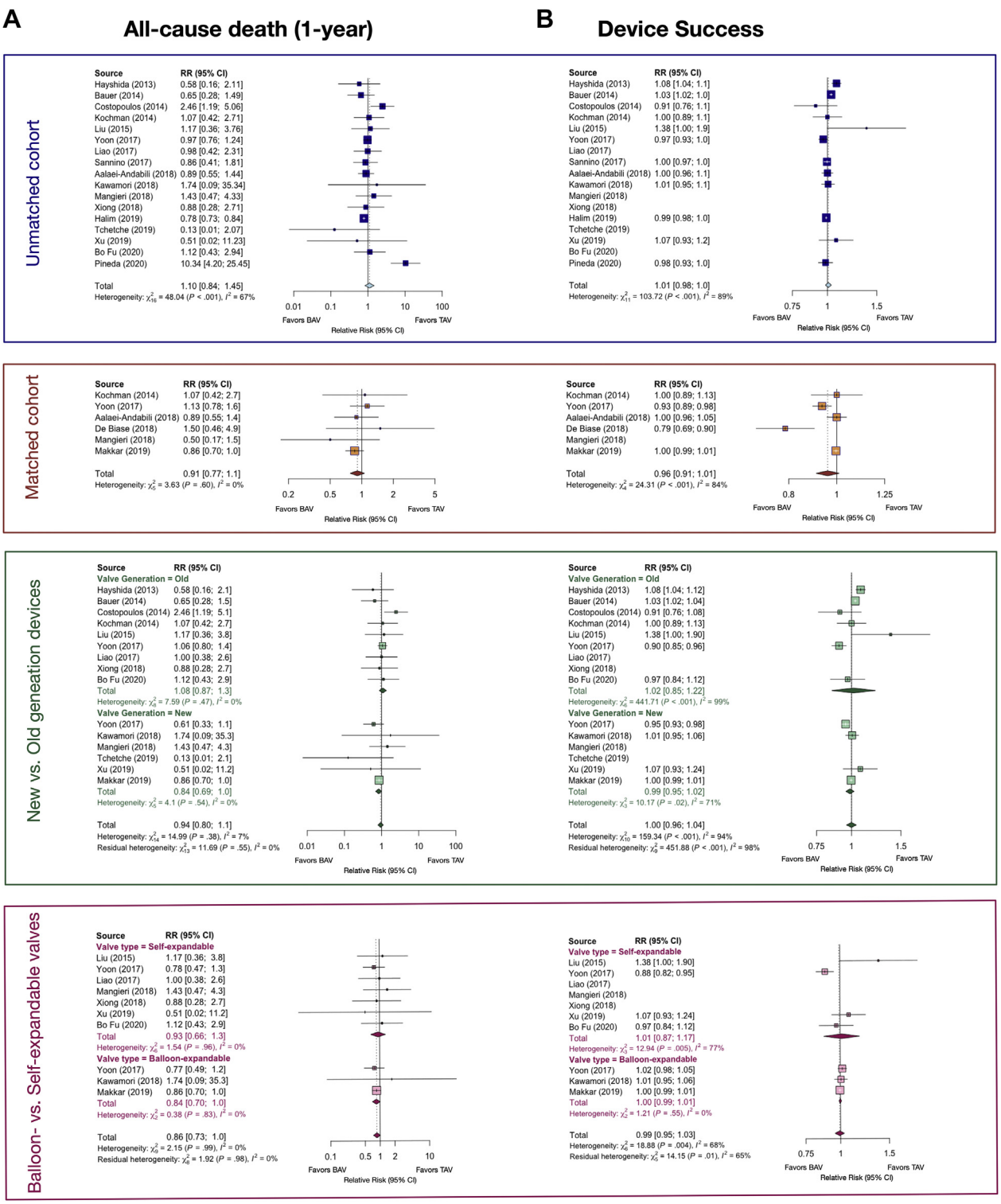
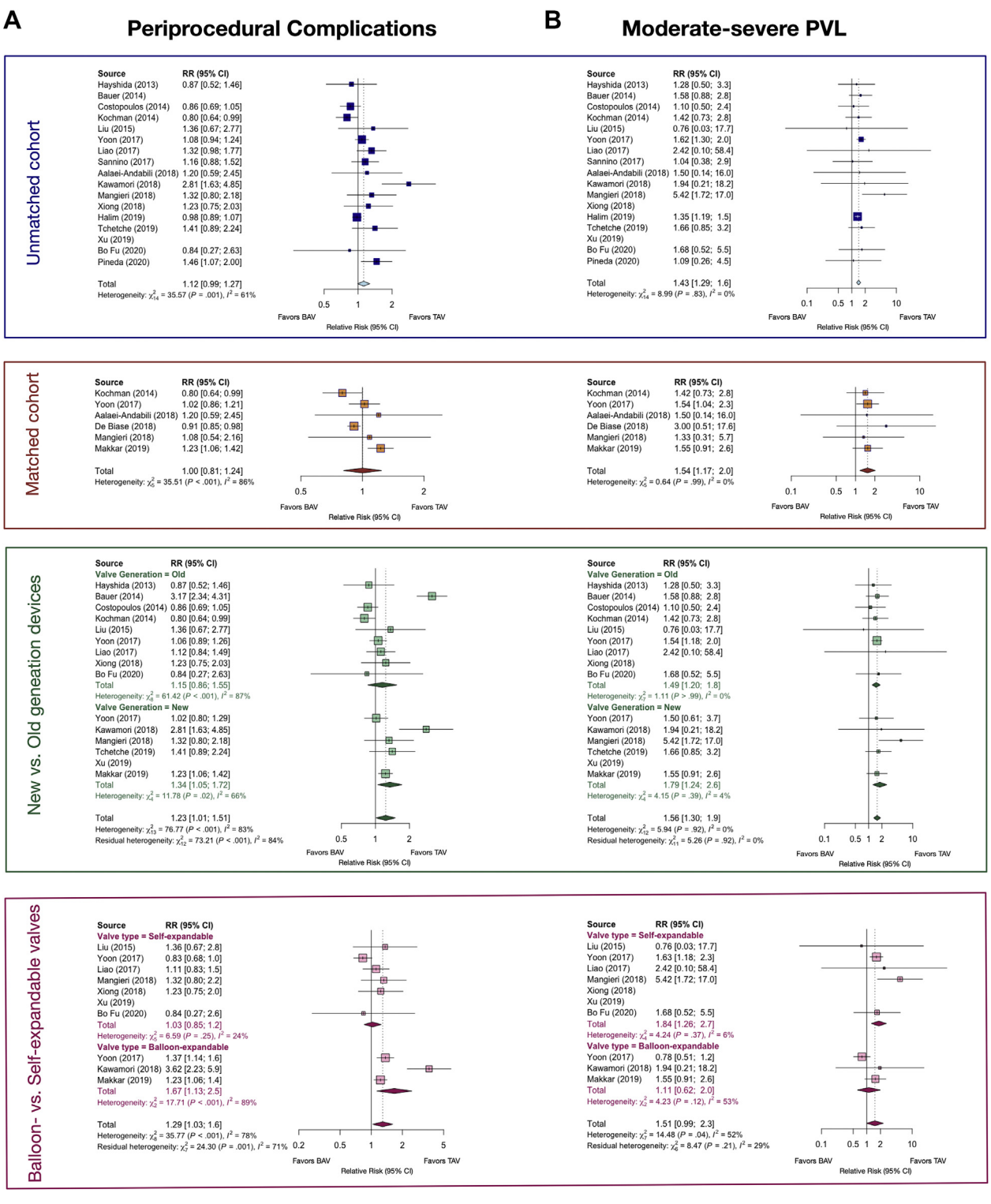
FIGURE 2 Forest Plots of Primary Endpoints



Relative risk (RR) of 1-year survival **(A)** and device success **(B)** is shown for each analysis. BAV = bicuspid aortic valve; TAV = tricuspid aortic valve.

FIGURE 3 Forest Plot of Secondary Endpoints



RR of periprocedural complications **(A)** and moderate to severe paravalvular leak **(B)** is shown for each analysis. Abbreviations as in Figure 2.

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TABLE 3 Incidence Rates of Individual Endpoints Explored in the Main Analysis and in the Matched Population

	Unmatched Population					Matched Population				
	BAV		TAV		*P* Value	BAV		TAV		*P* Value
Stroke/TIA	164	2.41 (2.03-2.78)	3,630	1.91 (1.39-2.43)	0.12	87	2.43 (1.91-2.95)	61	1.60 (1.19-2.01)	0.015
Annular rupture	2	1.07 (0.00-2.22)	5	0.15 (0.00-0.31)	0.12	7	0.27 (0.08-0.46)	0	0.02 (0.00-0.07)	0.014
Coronary occlusion	2	1.09 (0.00-2.31)	14	0.73 (0.09-1.37)	0.61	9	0.45 (0.00-1.08)	7	0.18 (0.03-0.32)	0.41
Device embolization	2	1.54 (0.00-3.29)	22	1.00 (0.23-1.78)	0.58	4	0.11 (0.00-0.24)	7	0.20 (0.03-0.36)	0.44
Conversion to surgery	54	0.79 (0.57-1.01)	990	0.56 (0.40-0.73)	0.11	36	1.20 (0.52-1.88)	12	0.35 (0.15-0.54)	0.018
Need for a second valve	170	3.74 (2.10-5.38)	2,193	1.36 (0.82-1.90)	0.007	48	2.87 (0.31-5.43)	21	0.94 (0.13-1.76)	0.16
Permanent pacemaker implantation	264	20.1 (13.7-26.6)	1,548	16.3 (12.2-20.4)	0.32	349	12.6 (8.3-16.9)	339	12.4 (4.5-17.4)	0.96

Incidence rates and 95% CIs are shown as percentages. Cumulative number of events is shown for each group; when raw numbers were unavailable, they were calculated from reported summary statistics. Results and *P* values are from the random-effect analysis.

TIA = transient ischemic attack; other abbreviations as in Table 1.

vs 90.8%; $P = 0.22$) in BAV and TAV and similar post-TAVR mean gradients (10.6 mm Hg vs 9.8 mm Hg; $P = 0.12$) and valve area (1.8 cm^2 vs 1.8 cm^2; $P = 0.86$), supporting a role for TAVR in these patients. These findings were also confirmed in our matched analysis on 7,071 subjects (Table 2).

Despite this, our study demonstrates that performing TAVR in BAV resulted in higher rates of individual complications such as conversion to surgery and need for a second valve, but the rate of periprocedural complications (RR: 1.87; 95% CI: 1.19-2.89) was similar to that observed in TAV in a cohort with similar baseline characteristics. Of note, this composite endpoint also included permanent pacemaker implantation, which has recently been added to the early safety endpoint in the Valve Academic Research Consortium-3 consensus (37). Despite being less relevant in terms of prognosis, this complication might be particularly relevant for subjects with BAV, who are generally younger.

Bicuspid anatomy is often accompanied by ascending aorta pathology, heavy calcification, and raphes, accounting for an intrinsic fragility of the aortic structures and higher rates of complications (38). In fact, the incidence of stroke or TIA and of annular rupture was significantly higher in BAV, despite the similar baseline characteristics of our matched cohort (Table 3). The higher incidence of stroke or TIA in BAV corroborates recent findings (39) and favors the use of embolic protection devices in this selected population. The incidence of annular rupture was exceedingly rare also in the BAV group (incidence rate: 0.3% vs 0.02% in TAV).

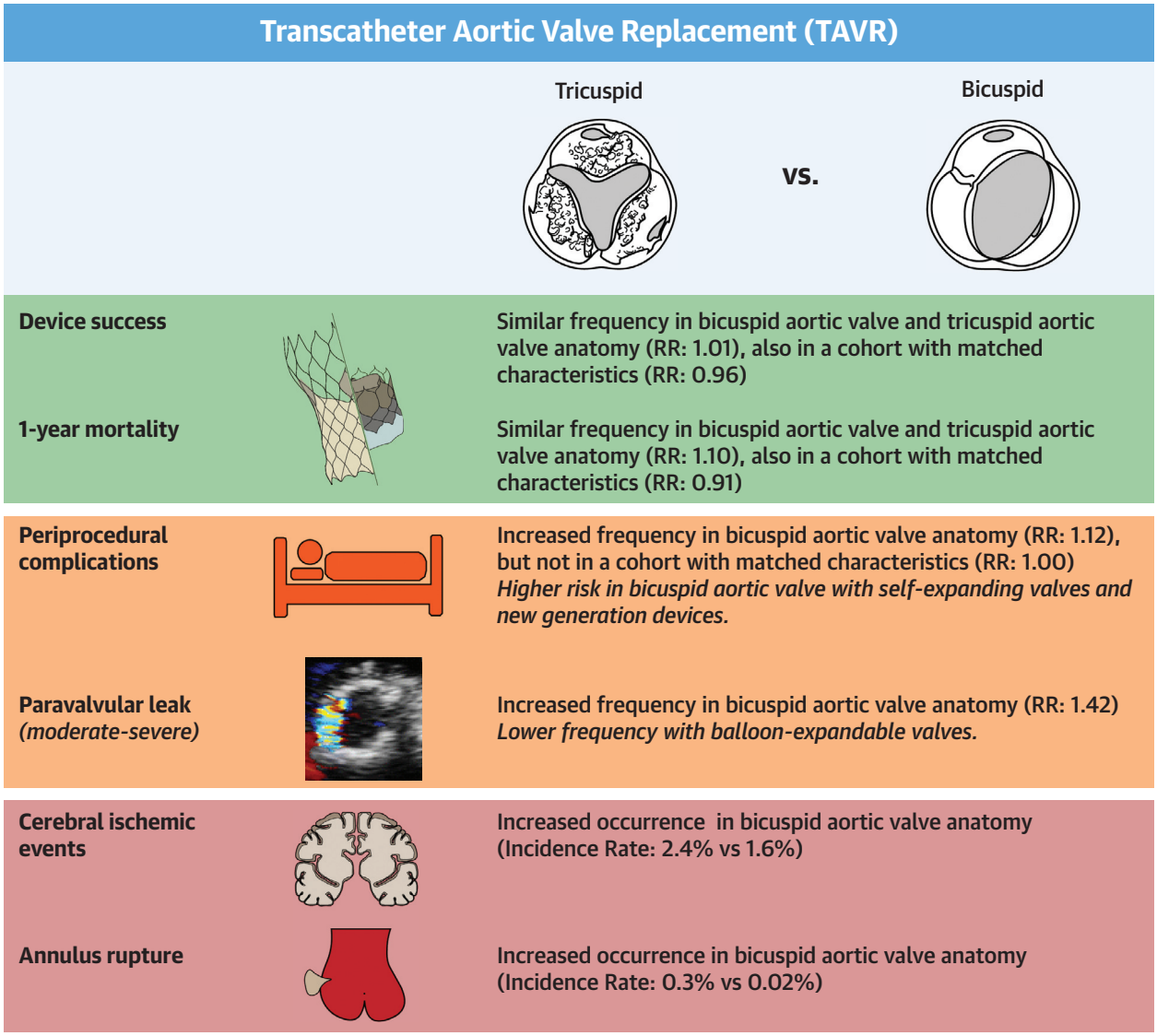
We additionally performed 2 separate subgroup analyses to explore the impact of device generation and valve type on outcomes. All-cause death and device success were similar in balloon- and self-expandable valves, but in subjects with BAV, a significantly higher rate of periprocedural complications was observed with balloon-expandable valves (RRs: 1.67 and 1.82, respectively) confirming previous findings from an international registry (36).

Interestingly, higher rates of periprocedural complications were also observed with new-generation devices. This result is not surprising, as new valve designs, which offer better performance, could have made operators confident to undertake TAVR in BAV, including patients and anatomies that were excluded in the past.

Finally, our meta-analysis confirmed that in BAV, TAVR performs poorly with respect to residual significant PVL. However, this finding was less pronounced in patients treated with balloon-expandable valves, as reported in retrospective registries (35,36). One explanation is that balloon-expandable valves, by exerting a higher radial force, may be able to circularize the often heavily calcified and elliptical BAV annulus, resulting in optimal sealing and lower PVL at the price of a higher rate of complications. Considering the relatively younger age of patients with BAV, it is mandatory to minimize PVL, and the designers of future generations of transcatheter valves should seek to address this technical challenge. Moreover, as both balloon-expandable and self-expandable valve technologies were associated with a distinct risk profile in subjects with BAV (higher periprocedural complications and significant PVL, respectively), dedicated technological advancement should be encouraged.

In conclusion, although BAV should not be a contraindication to TAVR and can be used safely in selected subjects with bicuspid anatomy with comparable long-term survival, the higher rates of periprocedural complications and of individual complication endpoints, such as cerebral ischemic

CENTRAL ILLUSTRATION Strengths and Weaknesses of Transcatheter Aortic Valve Replacement in Bicuspid Aortic Anatomy



Montalto, C. et al. J Am Coll Cardiol Intv. 2021;14(19):2144–2155.

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events and annular rupture, warrant expert operators at high-volume centers who can manage them promptly. The heart team should take into account these shortcomings of TAVR in BAV in the decision-making process.

STUDY LIMITATIONS. First, only nonrandomized, observational studies were included in this analysis, which limits the generalizability of our results. Although our sensitivity analysis including only a propensity-matched population accounted for heterogeneity and bias in baseline characteristics, residual bias cannot be excluded.

Second, studies reported the incidence of clinical endpoints without censoring for the occurrence of multiple events; therefore, our secondary composite endpoint should not be interpreted as the net of complications. In this regard, Supplemental Figure 2 offers insights into individual endpoint occurrence.

Third, PVL was evaluated in a different time frame from TAVR. Fourth, annular rupture was overall

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infrequent, and this might limit the generalizability of our findings. Moreover, there was heterogeneity in the reported outcomes across the various studies, so results should be interpreted with caution.

Finally, our BAV cohort consists of patients who were deemed unsuitable for surgical aortic valve replacement, which may result in a selection bias that is not accounted for by our statistical analysis.

CONCLUSIONS

In a large meta-analysis of 181,433 subjects, TAVR in BAV was associated with similar 1-year survival rates and device success as TAV. Nonetheless, the higher rates of moderate to severe PVL, annular rupture, and cerebral ischemic events observed in the BAV group warrant caution and further evidence. Our findings suggest that randomized clinical trials are required to ascertain if a percutaneous strategy is preferrable over surgery to treat patients with bicuspid anatomy.

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PERSPECTIVES

WHAT IS KNOWN? Limited observational data exist supporting TAVR in the context of bicuspid anatomy when surgical correction is unfeasible.

WHAT IS NEW? Selected subjects with BAV can undergo TAVR with similar outcomes and success rates as those with tricuspid anatomy, but the higher rates of periprocedural complications warrant expert operators who can manage them promptly.

WHAT IS NEXT? Adequately powered randomized trials are needed to determine whether these findings from observational data are generalizable. Device technology improvements dedicated to bicuspid anatomy might improve the outcomes of TAVR in these subjects.

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KEY WORDS aortic stenosis, bicuspid aortic valve, heart team, TAVR, transcatheter aortic valve replacement

APPENDIX For supplemental methods, a table, and figures, please see the online version of this paper.